

Optimal institutional design for collective deterrence

Torgeir Aambø

Norwegian Defence Research Establishment (FFI)

Abstract

Collective deterrence in a coalition of states depends on two key elements: the institutional design of decision-making, formalized as a social choice function that aggregates member states' levels of resolve, and the strength of the coalition's available means of escalation. This introduces a tradeoff between credible deterrence and escalation control, which we formalize and study via a simple signalling model. We compute its perfect Bayesian equilibria, and study the optimal institutional designs for both large and small coalitions by computing a variety of binormal and empirical ROC curves.

“Convince your enemy that he will gain very little by attacking you, which will diminish their enthusiasm.”

– Sun Tzu, *The Art of War* [CITE](#)

1 Introduction

Since the dawn of time there have been wars, and for just as long, people have tried to figure out how to avoid them. Deterring a potential enemy from attacking requires them to believe that the costs of such an attack outweighs their perceived benefits. In this simplified manner there are two strategies, reduce the potential benefits or increase the costs. The former is often counter-productive, though has been successful for example with the burn land tactic during the second world war, [CITE](#) . The latter, however, requires that the attacker believes the consequences to be large, for example by believing that large scale retaliation would occur – so-called second strike capabilities.

These ideas form the baseline for *deterrence* as a military strategy: avoiding an attack by having sufficiently powerful means of guaranteed retaliation [CITE](#) .

In essence, deterrence relies on a push-pull between two

fundamental objectives that are, in effect, the strategic analogues of Type I and Type II errors [CITE](#) :

1. the resolve to use retaliatory means have to be sufficiently credible, and
2. one should limit the possibility of escalation spirals and accidental use.

The dynamic between these two mechanisms is especially potent for nuclear deterrence, where the consequences are potentially of an existential nature. For a classical outlook on these fundamental issues see [CITE](#) and [CITE](#) .

NATO is the only proper nuclear alliance – in the sense that of being a multinational alliance and not a bilateral agreements – where the deterrence capabilities of individual nations are also shared to allies. However, NATO does not own nuclear weapons as a collective entity, nor does it operate a joint nuclear command structure with shared launch authority. For example, U.S. weapons stationed in Europe remain under exclusive U.S. custody and control, and any nuclear mission requires authorization from the relevant national leaders in addition to political approval through NATO's Nuclear Planning Group (NPG). [CITE](#)

Hence, there is currently no coalition of states in the world that jointly own both the nuclear deterrence capabilities and the command structure to act upon threats, as the historical proposals for a NATO multinational nuclear force never materialized [CITE](#) . However, such coalitions might become more politically viable now that arms-control frameworks, such as the New START treaty [CITE](#) , are intensifying the European discussions about burden sharing and the possible future role of France or the UK in European nuclear deterrence.

Currently, of course, the Non-Proliferation Treaty [CITE](#) serves as a block to the creation of such nuclear alliances. However, there is room to break the treaty

in the case of existential threats, or, alternatively, one could consider other deterrence technologies not limited by such restrictions. In the advent of such coalition then, the additional problem of *collective* rather than *unilateral* decision-making arises to complicate the above two problems. The problem of *credible resolve* becomes a problem of how the coalition’s states aggregate their individual resolves into a joint decision, and the problem of *escalation control* becomes a problem of not allowing this joint decision to be too easily affected by individual potential errors. Hence this joint institutional decision-structure introduces a trade-off between the two fundamental problems.

We can then phrase the motivating question for the present paper as: How does the institutional design of a coalition affect the deterrence mechanism and its effectiveness?

Such aggregations of individual choices is in social choice theory measured by a *social choice function*, see [CITE](#) . To answer this question then, we develop a simple signalling game model, that is dependent on weighted threshold choice function f that aggregates the coalitions votes on retaliation in to a shared resolve. The model hence covers the three fundamental elements of deterrence: capability, credibility and communication [CITE](#) . The coalition has to have deterrence capabilities in order to make the adversary’s perceived costs large, the resolve to actually retaliate using these capabilities has to be credible for the adversary, and finally, the coalition has to communicate thresholds and consequences.

To understand the interactions between deterrence and institutional design, we characterize the model’s perfect Bayesian equilibria, and study the credibility-false positive frontier. By approximating the probability of retaliation by a normal distribution this allows us to study optimal thresholds for large coalitions via binormal Receiver Operating Characteristic (ROC) curves [CITE](#) . We also study optimal thresholds and voting rules for small alliances, such as the nordic countries, via associated empirical ROC curves.

2 Theoretical background

Our model exists as a combination of two much studied mathematical models: signalling games and social

choice functions. These are each ubiquitous in their own respective fields of deterrence theory and voting theory. Before putting forth our combined model, we briefly explain the pieces separately.

2.1 Signalling games

Signalling games are dynamic Bayesian games, first studied in relation to learning by Lewis [CITE](#) . For a more modern treatment see [CITE](#) , and for deterrence specific signalling theory see [CITE](#) .

In a signalling game the players have incomplete information about the initial situation, and the game unfolds sequentially over time, with actions known to the other players [CITE](#) . More precicely, one player – often called the sender – has some private information about the state of the world, an takes an observable action S , called the signal, after which another player – often called the reciever – updates their beliefs about the world and chooses an action.

In our case of deterrence, incomplete information is essential; the uncertainties around the opponents resolve lies at the heart of the mechanism making deterrence a viable strategy.

Define perfect Bayesian equilibria.

2.2 Social choice functions

Social choice functions are mathematical models for ranking a set of alternatives, often called *social states*, by their collective desirability – combining the independent preferences of the individual into a common “will” of the people. Early versions were studied by Condorcet and Llull, [CITE](#) and [CITE](#) , but the modern theory now in use is often attributed to Arrow’s work [CITE](#) .

Social choice functions can be either *cardinal* or *ordinal*, [CITE](#) . Ordinal functions only use the relative ordering of social states to compute the best preferred state according to some mathamatical rule. On the other hand, cardinal functions use specific values attached to each social state, and hence is able to compute not just which state is preferred, but how much it is preferred to the other states. It is the former, the ordinal choice functions, that are most often used in public voting and electoral systems, and also the type of function

we will use in this paper.

2.2.1 Threshold rules

The most frequently used voting rule is the *majority rule*, stating the the preferred choice between two options is the one chosen by more than half of the voters [CITE](#) . In mathematical terms,

$$f(v; w) = \begin{cases} 1, & \sum_{i=1}^n v_i > \frac{n}{2} \\ 0, & \text{else.} \end{cases}$$

By varying the threshold for the rule outputting a 1, we also vary the choice function. For example, the consensus rule, used for example in NATO [CITE](#) , states that the preferred choice between two options is the one chosen by *all* voters:

$$f(v; w) = \begin{cases} 1, & \sum_{i=1}^n v_i = n \\ 0, & \text{else.} \end{cases}$$

Similarly, chosing a threshold $\frac{2}{3}$ gives the qualified majority rule and chosing $\frac{3}{4}$ gives the *supermajority rule*.

3 The model

Our model adopts this basic structure of a signalling game, as presented above, where we assume one of the players to be a coalition consisting of multiple states $\mathcal{C} = \{1, 2, \dots, n\}$. The other player is some potential attacking adversarial actor $\mathcal{A}$. Before we explain the individual parts, we summary the sequence of events:

1. the coalition has a predetermined fixed social choice function f , which is publically known;
2. nature draws aversary type T ;
3. nature draws signals s_i for each coalition member;
4. adversary actor $\mathcal{A}$ chooses to attack or not based on the above information;
5. in the case of an attack, the coalition votes on retaliation using f .

3.1 Types and signals

The adversary $\mathcal{A}$ has a *type*, $T \in \{0, 1\}$, where 1 represents being aggressive – actively planning an attack –

and 0 represents a benign adversary. More precisely, this is supposed to signify the intent to attack; if $T = 1$ then $\mathcal{A}$ intends to attack unless deterred, and if $T = 0$, then $\mathcal{A}$ prefers peace, but might still launch small probing attacks if the probability of retaliation is low enough. The coalition has a shared prior belief $\pi = \Pr(T = 1)$.

Each member i of the coalition $\mathcal{C}$ has a binary signal $s_i \in \{0, 1\}$, interpreted as its private stance on whether to retaliate or not, given that $\mathcal{A}$ attacks. This is dependent on their belief of the type of $\mathcal{A}$, giving a probability

$$p_i = \Pr(s_i = 1 \mid T = 1)$$

for whether coalition member i will vote to retaliate or not. Similarly, coalition members might recieve false alarms, which gives a probability

$$q_i = \Pr(s_i = 1 \mid T = 0)$$

of a retaliatory strike based on erroneous information. We assume that signals and probabilities are independent across members.

In the case of an attack, each coalition member $i \in \mathcal{C}$ then decides on a binary vote $v_i \in \{0, 1\}$, interpreted as their vote towards authorizing retaliation. In order to focus this paper on the institutional mechanisms, and not on whether the coalition members try to strategize for their own gain, we assume that $v_i = s_i$, which means that we assume the actors to vote truthfully based on their own intelligence assessments and internal stance.

3.2 Weighted threshold functions

The coalition is assumed to have a publically known decision rule, formalized by a social choice function f . In order to be able to analyze the behaviour of our signalling game with respect to such a function, we need to restrict to a class of viable rules. First, we assume a binary outcome, 1 representing coalition retaliation and 0 representing no retaliation. Second, we assume the choice function f to be a weighted threshold rule. This means that each coalition member i is assigned a weight $w_i \in \mathbb{R}$ dependent on the context of the attack, and that there is a threshold $\tau \in \mathbb{R}$, such that the aggregated vote of the coalition is 1 if the weighted sum of the individual members votes exceeds the threshold. This weight scheme is meant to include the possibility

that the attacked country in the coalition gets a vote that counts more towards the total, or that frontline countries have a larger say, or that a country has far better ISR capabilities etc.

Formally, then, given a list of votes coalition individual votes $v = (v_1, v_2, \dots, v_n)$, and a weight scheme $w = (w_1, \dots, w_n)$, we have

$$f(v; w) = \begin{cases} 1, & \sum_{i=1}^n w_i v_i \geq \tau \\ 0, & \text{else} \end{cases}$$

Most viable voting rules are formalizable in this framework, like the majority rule, consensus rule, dictatorship, qualified majority, veto players, etc. There are also some decision rules that are not possible to model in this framework, like multi-stage rules, lexicographical rules, non-monotonic rules and randomized rules. However, such rules are very unlikely to be viable for choosing a retaliation action, hence our focus on weighted threshold rules.

3.3 Retaliation probabilities

Assuming $\mathcal{A}$ attacks, retaliation will occur if the weighted number of coalition members receiving the signal meets the threshold. Because these signals are independent, the probability of retaliation is

$$R(\tau) = \Pr \left(\sum_{i=1}^n w_i v_i \geq \tau \mid T = 1 \right).$$

This function captures the *deterrence credibility* of the coalition $\mathcal{C}$. In other words it is the probability that an attack by the adversary $\mathcal{A}$ triggers a retaliatory response.

Similarly, if the adversary does *not* attack, then retaliation is accidental, with probability

$$F(\tau) = \Pr \left(\sum_{i=1}^n w_i v_i \geq \tau \mid T = 0 \right).$$

This function captures the coalition's *escalation risk*, i.e., the probability that the coalition accidentally authorizes a response when there had been no attack. This could happen if, for example, another actor than $\mathcal{A}$ was responsible for an attack on the coalition, or that an accident is interpreted as an attack.

Example 3.1. In the simple case that each coalition member i has an equal weight, say $w_i = 1$ for all i , as well as equal probabilities p_i , and the threshold τ is a positive integer less than or equal to n , then these probabilities become binomial, with

$$R(\tau) = \sum_{k=\tau}^n \binom{n}{k} p^k (1-p)^{n-k},$$

and similarly

$$F(\tau) = \sum_{k=\tau}^n \binom{n}{k} q^k (1-q)^{n-k}.$$

The often used majority rules and consensus rules are both of this nature.

3.4 Costs and benefits

The preferred outcome of the adversary actor $\mathcal{A}$ is to attack and suffer no consequences. The preferred outcome for the coalition $\mathcal{C}$ is that no attack occurs.

In order for an attack to be of interest, there has to be a perceived benefit for $\mathcal{A}$. In the situation of preferred outcome for an adversary $\mathcal{A}$ of aggressive type $T = 1$, where it attacks and no retaliation occurs, we denote this benefit by B_1 . We assume this value to be a number, for example monetary benefit or some other measure of benefit that $\mathcal{A}$ deems desirable. If retaliation occurs, then $\mathcal{A}$ suffers a cost C_1 , which we assume is of comparable type to the benefit B_1 .

Similarly, if the adversary has benign type $T = 0$, then there might still be a small benefit B_0 for initiating an attack, if the probability of retaliation is small enough. This tries to capture low-stakes provocations and probing actions. Similarly, there is a cost C_0 . These are assumed to be smaller than the payoffs for the aggressive type, giving inequalities

$$B_0 < B_1 \text{ and } C_0 < C_1.$$

We can then compute the expected payoffs for a adversary of a given type. The probability of retaliation by $\mathcal{C}$ is given as above by $R(\tau)$, hence the expected payoff for $\mathcal{A}$ with type $T = 1$ for attacking is

$$E_{T=1}(\text{attack}) = (1 - R(\tau))B_1 - R(\tau)C_1,$$

and similarly

$$E_{T=0}(\text{attack}) = (1 - R(\tau))B_0 - R(\tau)C_0.$$

3.5 The equilibria

As our model is a very simple from a game theoretical perspective, the equilibria are quite easy to find. In our model, only $\mathcal{A}$ is a strategic actor, who makes a choice of an action based on some information – the coalition $\mathcal{C}$ does not perform a strategic action, it is decided upon by their predetermined institutional design.

Assuming rationality, the adversary state attacks if it deems the expected payoff to be greater than 0. Rearranging the terms gives that $\mathcal{A}$ of type $T = 1$ attacks if and only if

$$R(\tau) < \frac{B_1}{B_1 + C_1},$$

and that $\mathcal{A}$ of type $T = 0$ attacks if and only if

$$R(\tau) < \frac{B_0}{B_0 + C_0}.$$

These fractions give the adversary actor $\mathcal{A}$'s *deterrence thresholds* and *deterrence intervals*. For simplicity, denote by $\bar{R}_1 := \frac{B_1}{B_1 + C_1}$ and $\bar{R}_0 = \frac{B_0}{B_0 + C_0}$. By the inequalities above, we get $\bar{R}_0 < \bar{R}_1$. This means we have three possibilities, which corresponds to the three possible equilibrium regions:

1. if $R(\tau) < \bar{R}_0$, then the adversary actor $\mathcal{A}$ attacks, no matter its type;
2. if $\bar{R}_0 < R(\tau) < \bar{R}_1$, then a benign adversary is deterred, but an aggressive one attacks;
3. if $R(\tau) > \bar{R}_1$, then the coalition $\mathcal{C}$'s collective deterrence has succeeded in deterring also an aggressive adversary.

The Bayesian updating of beliefs are trivially satisfied by the fact that the coalition is not a strategic actor in this game. We simply define the coalition's posterior belief $\Pr(T = 1 \mid \text{attack})$ to update according to Bayes rule.

These three strategies and beliefs constitute perfect Bayesian equilibria, as the coalition's behavior is determined by the weights w and the threshold τ , and the adversary's actions satisfies sequential rationality given its beliefs about the probability of retaliation $R(\tau)$. The first and last regions above means that the corresponding equilibria are both *pooling*, while the second is a *separating* equilibrium. They are also the only possible ones, and thus yields a full classification.

Remark 3.2. The costs C_1 and C_0 depends on the coalition $\mathcal{C}$'s available deterrence technology. For example, having strategic nuclear weapons significantly increases the costs of retaliation for the adversary $\mathcal{A}$. As discussed in the introduction, new nuclear alliances are prohibited by the Non-Proliferation Treaty [CITE](#). But, in certain situations and contexts, it is possible to circumvent this, or alternatively use other available or future technologies. We can then study two different costs – one with and one without sufficient deterrence technologies – and see how this affects the equilibria. Doing this reveals the natural outcome: larger costs means smaller expected payoffs, and larger intervals where deterrence is successful.

As the game is very simple, it is not the equilibria that are interesting. The interesting question about our setup is, in our opinion, trying to determine which social choice functions that yield the best deterrence properties for the coalition $\mathcal{C}$. In the next section we attempt to analyze this question.

4 Optimal thresholds

For a given coalition $\mathcal{C}$, there are countless possible choices for an institutional design f that computes the aggregated decision based on coalition members' votes. As mentioned earlier, this function f allways introduces a tradeoff between credible deterrence and escalation controll.

For $\mathcal{C}$ then, it is highly important to understand which such designs that give the best deterrence credibility while minimizing the risk of accidental retaliation. In essence, this is the classical tradeoff between the probability of *true positives* and the probability of *false positives*. There is a rich history from many fields of science studying this exact tradeoff, see for example [CITE](#) and [CITE](#), that we can use to study this issue in our setting.

Our choice function f has two possible outcomes, retaliation or no retaliation, based on two possible situations of true or false attacks. By false attack, or no attack, we here mean that there has been a false detection due to an accident, or an attack that is misattributed to the wrong adversary. This means that the function f is a binary classifier, in the sense of [CITE](#).

The standard way of measuring the accuracy of a binary classifier is to measure the correct and incorrect assignments. For this we have a simple contingency matrix:

	Retaliation	No retaliation
Attack	True positive	False negative
No attack	False positive	True negative

This gives us the *true positive rate* (TPR), defined as

$$\text{TPR} = \frac{\text{true positive}}{\text{true positive} + \text{false negative}}$$

and the *false positive rate* (FPR), defined as

$$\text{FPR} = \frac{\text{false negative}}{\text{true positive} + \text{false negative}}.$$

These are, respectively, the probability that retaliation occurs, and the probability of an accidental retaliation, which we have previously named $R(\tau)$ and $F(\tau)$.

The binary classifier is dependent on the threshold τ , which allows us to visualize the classifiers performance on the true positive-false positive tradeoff by ROC curves.

4.1 ROC curves

Receiver Operating Characteristic (ROC) curves were first used for radar signals during World War II, where they were used as an analysis tool for strengthening the predicting power of radars after the Pearl Harbor attack [CITE](#). Since then they have found a wide use throughout science, mainly in medicine [CITE](#), epidemiology [CITE](#), and more recently as a tool for evaluating classification algorithms in machine learning [CITE](#) and artificial intelligence [CITE](#).

The ROC space is defined by having TPR and FPR as its axes, and as these are rates from 0 to 1 the space is equivalent to the unit square. Any point in this space represents a tradeoff between true positives and false positives. The point $(0, 1)$ is the *perfect classifier*, in the sense that it correctly identifies 100%, with 0% errors.

By plotting $(F(\tau), R(\tau))$ for all different thresholds τ , one gets a curve through ROC space that visualizes the effect of the threshold on this tradeoff. The point on

this curve that is closest to the perfect classifier $(0, 1)$ is the optimal threshold for the given choice function.

By making ROC curves for a variety of choice functions, we can then identify the *best* institutional design for the coalition $\mathcal{C}$, in terms of maximizing correct retaliations while minimizing accidental use.

We do this in two different situations: a large coalition and a small coalition. For specificity and to relate this paper to ongoing discussions in Norway, we study the Nordic countries as our small coalition. The small number of members allow us to compute the ROC curves empirically for a variety of choice functions f . For large coalitions, however, we use a normal approximation to the probabilities, and plot binormal ROC curves, as in [CITE](#), instead.

4.2 The Nordics

The ROC curves are dependent on the probabilities of correct identification and the coalition members belief about the type T of the adversary state $\mathcal{A}$. In order to compare choice functions based on their performance, we plot choice functions based on several natural weight schemes $w = [w_1, w_2, w_3, w_4]$ at the same time – for fixed probability vectors $p = [p_1, p_2, p_3, p_4]$ and $q = [q_1, q_2, q_3, q_4]$.

To have a couple base lines, we look at the case where all nordic states have the same probability of voting in favor of retaliation given the type of $\mathcal{A}$. The following shows the ROC curves for varying weight schemes when both the shared Nordic resolve and the accuracy for correct identification are *high*.

4.2.1 Border states and intelligence

4.2.2 Veto rules

4.3 Large coalitions

5 Conclusion

